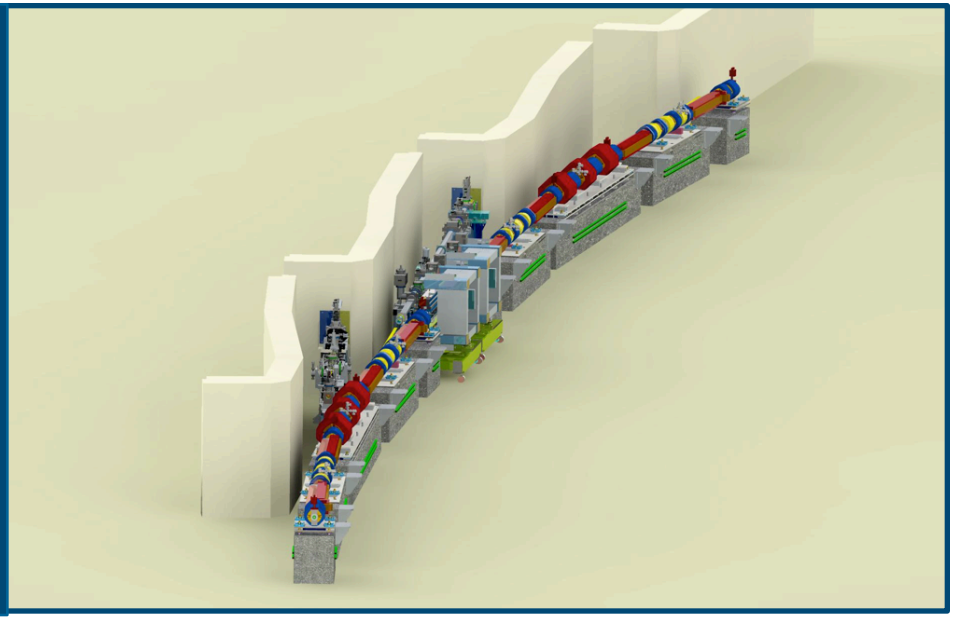
APS Upgrade Plans
(and what the CATs should be doing to prepare)



Dennis Mills
APS Deputy Director

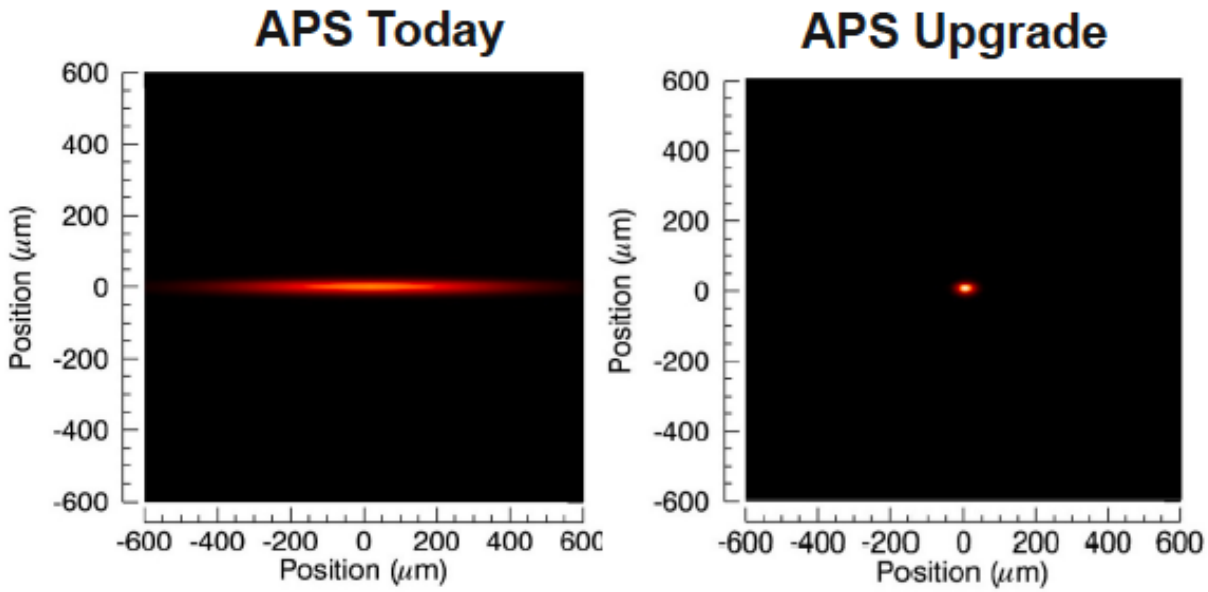
LS CAT Board Meeting
December 8, 2016

The APS Upgrade: Building the world's leading high brightness hard x-ray source

The APS Upgrade is a **next-generation** facility:

- Optimized for hard x-rays
- Incorporating advanced beamlines, optics and detectors
- 'Round' source ideal for imaging

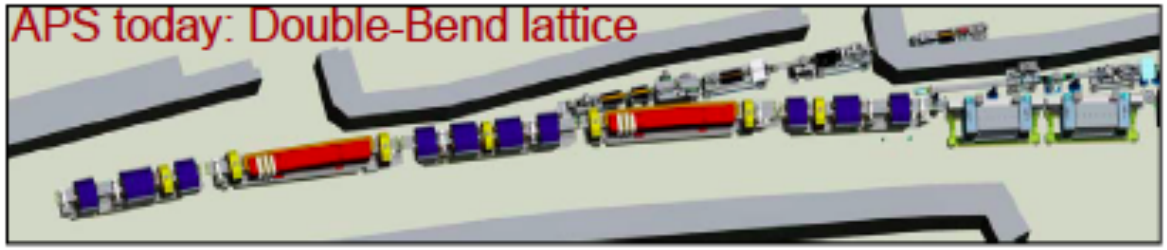
Transverse electron (source) size

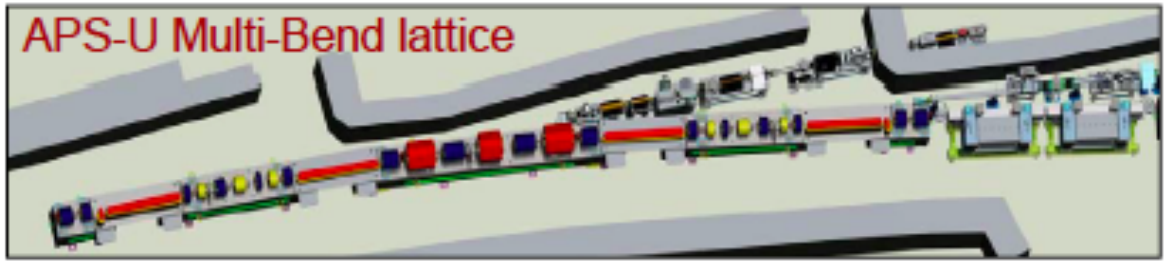


APS-U exceeds the capabilities of today's storage rings by **2 to 3 orders of magnitude** in brightness, coherent flux, nano-focused flux

APS-U design concept

~50-fold reduction in horizontal emittance





$$\varepsilon_x = C_L \frac{E^2}{N_D^3}$$

E = Beam energy
N_D = Number of dipoles per sector

– J. Murphy, BNL-42333

APS Upgrade Features

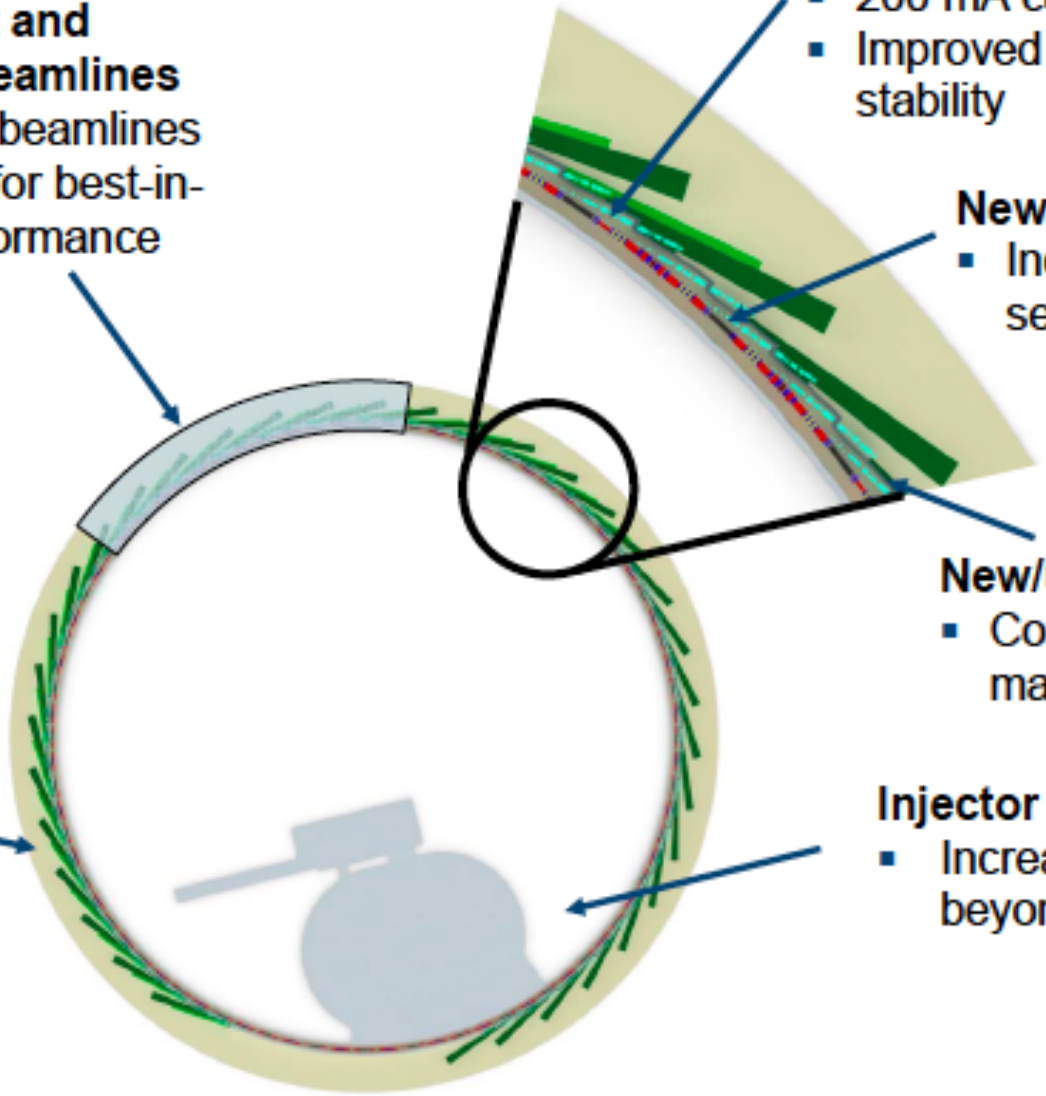
- 4th generation storage ring based on multi-bend achromat lattice
- Design for high-brightness, ultra-low emittance: $\varepsilon_x < 75$ pm goal (objective KPP)
- Diffraction limited vertical emittance to 15 keV, horizontal emittance to 2 keV
- Flexible operation: High-brightness and timing modes, round and flat beams
- Reuse existing infrastructure valued at $1.5B
- World leading experimental capabilities with a suite of new/rebuilt/heavily-upgraded state-of-the-art beamlines included in the project
- 35 ID straight sections with full suite of ~69 operating beamlines in APS-U era
- One-year dark period is a key project deliverable

APS Upgrade Project Scope

New storage ring
- 6 GeV MBA lattice
- 200 mA current
- Improved electron/photon stability

New, rebuilt and upgraded beamlines
- Suite of 8 beamlines designed for best-in-class performance

New insertion devices
- Incorporate SCUs on selected beamlines



Beamline Enhancements:
- Improvements to make all beamlines 'Upgrade Ready'

New/upgraded front-ends
- Common design for maximum flexibility

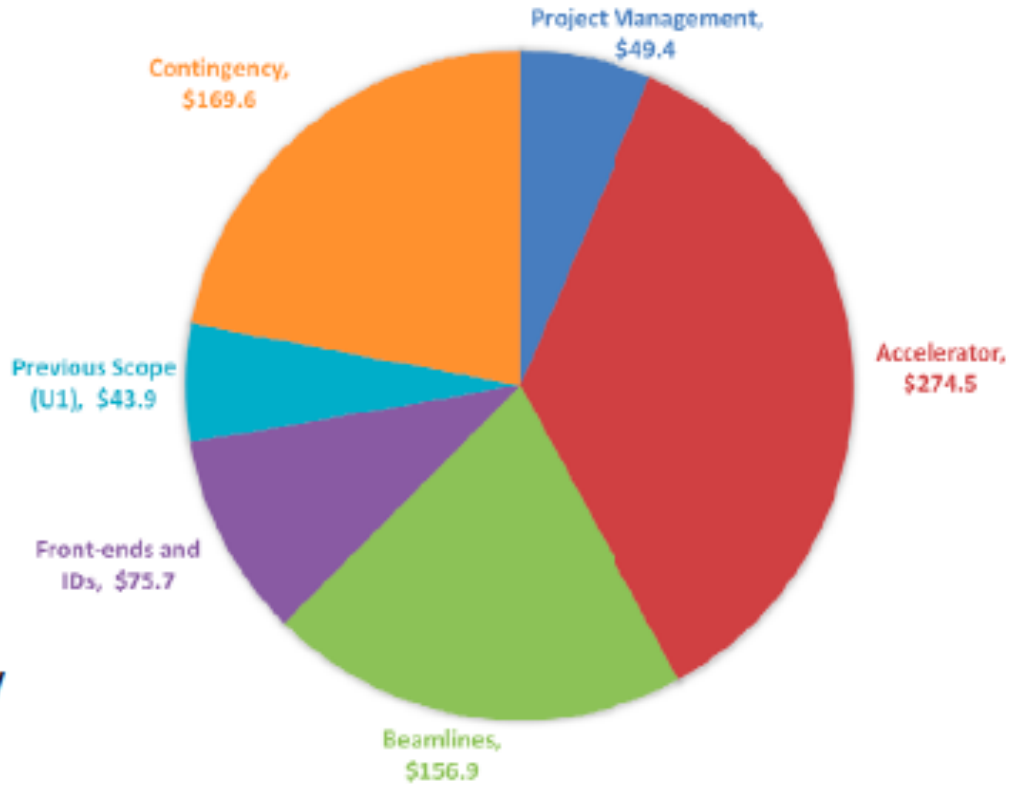
Injector improvements
- Increase performance beyond present capability

APS-U cost and funding profile

APS-U TPC is $770M, including 33.5% contingency on work to go

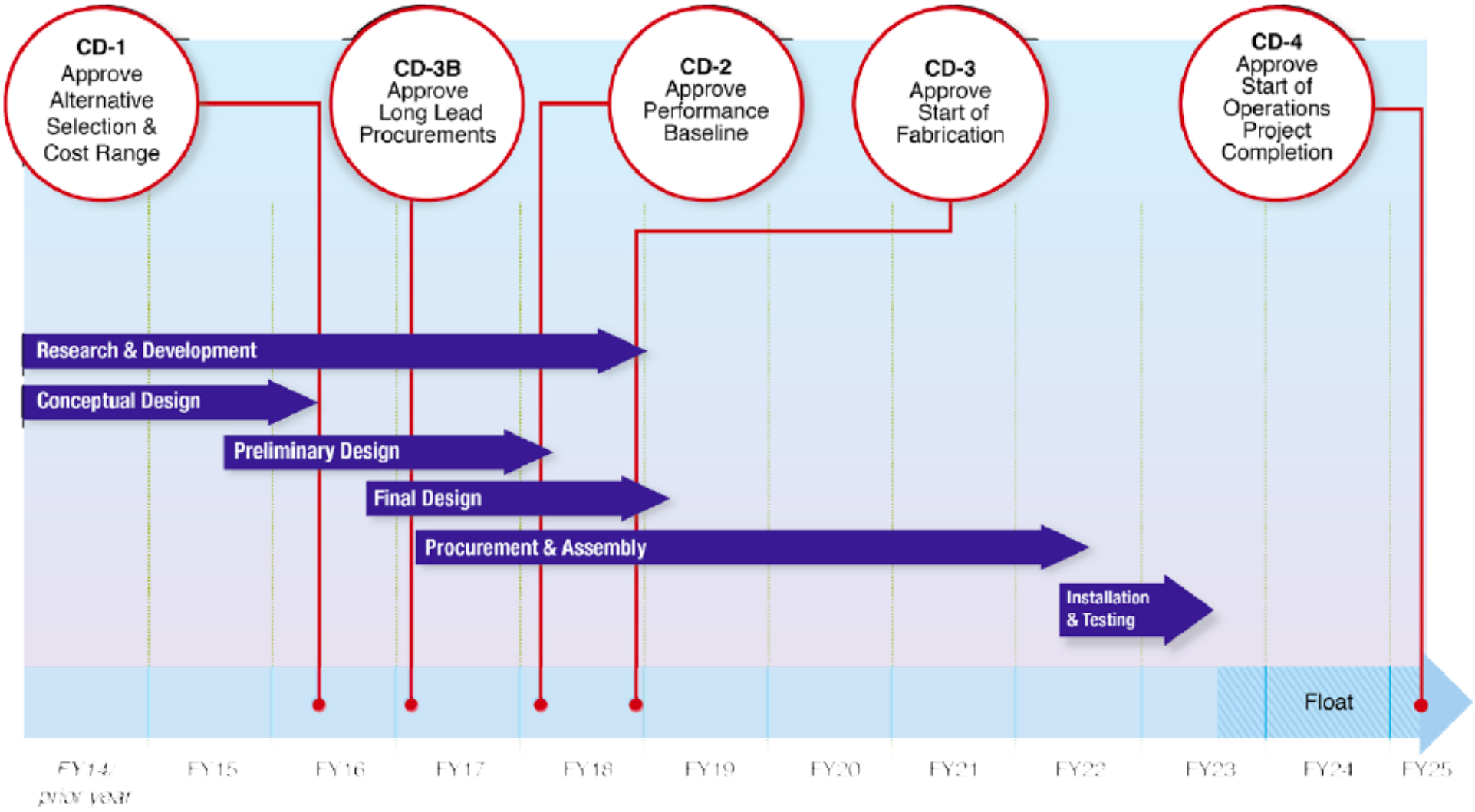
TPC includes costs associated with previous version of Upgrade

- APS-U has been funded at $20M/year for FY12-16
  - Costed/committed ~$95M to date
- Administration's FY17 Budget Request is $20M
  - Impact is $30M and one-year delay



Funding (BA) in Millions of Dollars (Then-year Dollars)												
	All Prior Years	FY 2014	FY 2015	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	FY 2023	Total
OPC	8.5								5	27.5	10	51
TEC	40	20	20	20	20	46.7	93.3	160	185	114		719
Total	48.5	20	20	20	20	46.7	93.3	160	190	141.5	10	770

APS Upgrade Project Schedule



This schedule is based on proposed funding profile.
From a technical point of view the project is ready to proceed more rapidly.

Argonne NATIONAL LABORATORY

Beamline Baseline Enhancements or "Do No Harm Policy"

- As part of APS-U "do no harm" policy, all beamlines are eligible for
  1. new high heat load front end (if necessary) and an ID providing comparable performance as existing ID (***funded out of FE/ID Budget of APS-U***)
  2. realignment of BM beamlines to compensate for source point movement in new lattice (***funded out of the Beamline Enhancement budget)***
  3. new/refurbished optics and other hardware to ensure that the beamline performs as well as it currently does (***funded out of the Beamline Enhancement budget)***



Primum non nocere

Further (Discretionary) Enhancements

- A questionnaire was sent out requesting beamline personnel to list possible enhancements aimed at taking advantage of the MBA source properties.

- Based, in part, on that questionnaire a budget of $25M was put forth for beamline enhancements (does not include overhead).

- Items #2 and #3 on previous page are commitments we have made to XSD and CAT beamlines and the funding, which comes out of the beamline enhancement budget, is “non-discretionary”.
  - The project estimates that these “non-discretionary” costs will be less that $5M, so the “discretionary” budget for beamline enhancements is over $20M.

Allocation Process for Discretionary Enhancements

- Call for 1-2 page proposal describing the scientific case for enhancement funds, community it would support, and what funding would it would be used for
  - Enhanced sources (SCUs, revolvers, 3-pole wigglers)
  - Upgraded optics (monos, mirrors, etc.)
  - Vibration isolation optical components and/or sample environments
  - End station instrumentation and computer hardware

- Upgrade estimates the cost of the enhancements for each proposal.

- Proposals are reviewed and prioritized by ad hoc committee - primarly APS/APS-U staff/managers, but would include CAT/PUC rep(s), member(s) of ESAC, etc.

- Prioritized list presented to SAC at March 2017 meeting **OR** via teleconference (at a later date) for advice and comment.

Review Criteria and Timeline

- Review Criteria
  - Scientific impact
  - Degree of benefit to the General User Program (beamlines that allocate a larger portion of time for access by General Users favored over those that allocate a smaller portion of time)
  - Alignment with upgraded APS capabilities and APS strategic plan (high energy, coherence, etc.)
  - Bang for the buck

- Would like this process completed late-Spring/Summer of 2017 so that the proposed enhancements can included in the project plan before the CD-2 review (late-Summer/Fall of 2017).

- Notes:
  - Changes can be made, if necessary, after the CD-2 review
  - Enhancement funds could change (up or down) once the beamline roadmap is completed and better costs for the proposed full beamlines are determined
  - Some funds could be directed towards mitigating impact of long beamlines

What Should The CATs Be Doing?

- Developing a plan for your science program post APS-U.
  - Doing the same thing (PX) or something different (SAXS, imaging, serial PX, etc.)?
- Developing a plan for required beamline enhancements to meet your post-upgrade science program goals.
  - What hardware is needed?
  - Delivery times, vendor capabilities, etc.?
  - Do you want to make the hardware enhancements before the dark period?
- Developing a plan for the 1 year dark period (and the “dim” period likely to follow for several months).
  - Write papers
  - Funding and activity of BL staff
  - Access to other facilities (a work in progress by APS)




Questions…